STATEMENT OF WORK (SOW)

SOW No: 20260809001

Project Title: Multilogue Fluency and Multi-Word Unit (MWU) Research Processing System
Client: Hiroshima University
Provider: TrendX(Dalian)Technology Co., Ltd
Date: August 9, 2026

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1. Executive Summary

This Statement of Work defines the modular implementation of a research-support pipeline for L2 multilogue fluency and multi-word vocabulary analysis. The system converts a confidential room-mix recording into traceable speaker-attribution evidence, dual-threshold Praat-compatible interaction drafts, reviewed transcript units, lexical and MWU feature tables, and a final research matrix. Automated outputs are pre-annotations only. Final research data remain the researcher-corrected TextGrid, transcript, and approved matrix.

Commercial structure: Layer 1 (L1a + L1b) is the recommended first contract because it addresses the most labor-intensive speaker and Praat annotation work. Layers 2 and 3 are optional and may be activated separately after the required research definitions and reference data are accepted.

2. Scope of Services

The project consists of four contracted modules mapped to the five-phase research workflow.

Layer 1a: Phase I — Shared Evidence, Speaker Attribution and Isolation

- **Inputs:** original room-mix WAV, task boundaries, speaker-count setting, canonical speaker IDs, and client-approved provider configuration.
- **Processing:** local audio preflight; canonical WAV clock; acoustic room-activity evidence; AI diarization and timed-word evidence; explicit provider-to-canonical speaker mapping; Stage-1 vocalisation / laughter / artifact / uncertainty evidence.
- **Outputs:** speaker activity RTTM/CSV, speaker TextGrid, provider evidence archive, confidence and review flags, one full-duration muted-mirror listening track per identified speaker, and a callable floor / nine-label engine for Phase II.
- **Scope boundary:** muted-mirror WAVs mute non-target intervals but are not source-separated clean-room recordings. The current six-tier acceptance baseline is a three-speaker task. Phase I speaker count is configurable; extending the interaction schema beyond three speakers requires an agreed N+3 validation sample.

Layer 1b: Phase II — Multi-Threshold Praat/TextGrid Automation

- **Inputs:** original room-mix WAV, Phase I evidence and engine, versioned method parameters, and the research-team review strategy.

- **Independent runs:** P = 0.25 s and P = 0.35 s are executed separately on the same canonical timeline. The threshold list remains configurable. The 200 s intensity window and Scale Times/full-timeline behavior are recorded in the method manifest.
- **Baseline route:** Path B. Qualified or sub-threshold overlap at a transition is reported as present with offset not measured; it is never serialized as zero. Underlying turn-end, turn-start, raw-gap, overlap, provider-capability and evidence IDs are retained for future Path A recomputation. Path A is not a parallel deliverable; if researcher-reference validation supports it, switching from Path B to Path A using the stored evidence is included and is not a CR.
- **Outputs per threshold:** one six-tier TextGrid; nine-label interval table; floor timeline; transition evidence; flags; interaction summary; duration and per-pause tables; overlap-capability evidence; method/provenance manifest; validation report.
- **Downstream handoff:** the reviewed nine-label timeline feeds Phase IV. Signed FTO, where supported, feeds Phase V only and is not used to determine Phase IV lexical/MWU features.
- **Human gate:** the research team corrects the selected draft(s) in Praat. The review strategy—both thresholds manually reviewed, or a primary threshold reviewed and the secondary regenerated deterministically—must be confirmed in writing before Layer 1 acceptance is finalized.

Layer 2: Phases III–IV and Early Phase V — Research Analysis Expansion

- **Purpose:** produce RAW-TIMING and TIDY-PHRASE transcripts, AS-unit/clause mappings, lexical and MWU feature tables, label-aware pause-location features, and rate metrics.
- **Start conditions:** researcher-approved transcript conventions, AS-unit/clause rules, MWU operational definition, syllable and repair coding rules, tool/version settings, and representative gold examples.
- **Alignment boundary:** base scope uses reviewed silence plus researcher-approved clause/AS-unit boundaries. Word-level forced alignment is an optional, separately validated add-on; no word-level timing claim is made without alignment evidence.
- **Risk condition:** TAALES, TAALED, AntConc, MWU and pause-location outputs depend on third-party tool behavior and client-supplied operational definitions. If those inputs are incomplete, the parties may reduce, defer, or substitute the affected deliverables by written agreement.

Layer 3: Phase V — Final Matrix, Validation, Reporting and UI Completion

- **Purpose:** merge accepted Layer 1 and Layer 2 outputs into an R/Python-ready matrix, research workbook, codebook, validation package, method logs, archive-ready export, and finalized operational Web UI.
- **Start conditions:** accepted upstream artifacts, signed final matrix schema, field-level codebook, expected sample rows, validation metrics, and reporting format.
- **Risk condition:** Layer 3 cannot guarantee statistical findings, publication acceptance, or valid synthesis when upstream definitions or gold/reference data are absent. The final deliverable set may be coordinated to the available validated inputs before Layer 3 activation.

Research Integrity and Data Status

- AI diarization, ASR, local VAD and rule-based labels are draft evidence and must not be represented as human-verified data.
- The original WAV is the canonical acoustic clock. Praat review and researcher correction establish the final TextGrid used for analysis.
- Every production run records method parameters, provider/model identity where available, timestamps, checksums, review state and export lineage.

3. List of Deliverables

The Provider shall deliver the following items for each activated layer. Unactivated optional layers are not included in the contracted deliverable set.

A. Layer 1a Deliverables

1. Speaker-attribution evidence package: RTTM, CSV/JSON, provider mapping, confidence and review flags.
2. Speaker TextGrid and full-duration muted-mirror WAVs for researcher listening and correction.
3. Phase I manifest, canonical-clock report, security/provenance log and validation summary.

B. Layer 1b Deliverables

1. P025 and P035 Praat-compatible TextGrids using the approved tier and label schema.
2. Nine-label intervals, duration summary, per-pause table, interaction summary, floor timeline, transition evidence and review flags.
3. Overlap-capability evidence, method manifest, deterministic replay record and client review package.

C. Layer 2 Deliverables (Optional)

1. RAW-TIMING and TIDY-PHRASE files, unresolved-boundary report, and approved AS-

unit/clause map.

2. TAALES/TAALED/AntConc-compatible lexical outputs, MWU tables, pause-location features and rate metrics, subject to the signed definitions.
3. Early Phase V merge table and method/validation notes for Layer 2 features.

D. Layer 3 Deliverables (Optional)

1. Final R/Python-ready matrix, research workbook and data codebook.
2. Validation report, gold/reference comparison package, method log and archive manifest.
3. Finalized Web UI, downloadable artifact browser and deployment/user documentation.

E. Technical Handover

- Versioned source code and deployment configuration for activated modules.
- User operation guide and system architecture/workflow document.
- No participant transcript text, credentials, signed URLs or absolute client paths will appear in public documentation or software logs.

4. Technical Environment

- **Application:** browser-based Web UI with background processing workers; deployable on an agreed Linux VPS and testable on macOS development systems.
- **Audio/Praat toolchain:** WAV/PCM master audio, FFmpeg/ffprobe, Praat CLI/headless scripts, TextGrid, CSV, JSON and XLSX outputs.
- **AI evidence:** initial provider baseline may use pyannoteAI Community-1 for diarization and AssemblyAI Universal-3-Pro for timed transcript evidence with disfluencies enabled. Exact provider/model versions are recorded per run and may be replaced only with written approval or documented equivalence testing.
- **Security:** client research audio is confidential. It shall not be reshared or sent to an unapproved third party. Provider/API use and retention settings must be approved before production processing.
- **Canonical timeline:** local WAV duration measured by ffprobe; external timestamps are clipped/rejected against this timeline and never extend it.

5. Project Timeline (Estimated)

- **Layer 1 (L1a + L1b):** approximately 4–6 weeks after kickoff, complete inputs, review-strategy confirmation and validation-set availability.
- **Layer 2:** approximately 2–4 weeks after all research definitions, reference examples and Layer 1 reviewed artifacts are accepted.
- **Layer 3:** approximately 1–2 weeks after Layer 2 acceptance and final matrix/codebook sign-off.

Schedule condition: these estimates exclude client/researcher review time, institutional approval delays, unavailable third-party services, and changes to the frozen methodology or deliverable structure. A dated delivery plan will be issued for each activated layer at kickoff.

6. Out of Scope (Exclusions)

Unless separately activated or agreed in writing, the following are excluded:

- Human expert annotation, full manual transcription, rater recruitment, and researcher correction labor.
- Guaranteed source separation from a single-room mix; muted-mirror tracks are review aids, not isolated clean speech stems.
- Guaranteed automatic accuracy, zero-edit TextGrids, statistical conclusions, research interpretation, publication writing, or publication acceptance.
- Creation of missing AS-unit, clause, MWU, syllable, repair, overlap or independent validation gold standards.
- Word-level forced alignment/MFA, phoneme-level alignment, or manual correction of aligned boundaries unless separately quoted.
- Institutional ethics approval, participant consent administration, external software licenses, or unapproved third-party API charges.
- Major changes to the agreed methodology, architecture, data schema or deliverable structure after scope freeze.

Included tuning: routine adjustment of P values, minimum sounding runs, minimum overlap, gap filling, and comparable calibration parameters is included and does not trigger a Change Request.

7. Commercial Terms

Item	Scope	Description	Price
Layer 1	L1a + L1b	AI speaker evidence/isolation + dual-threshold Praat/TextGrid automation	USD 2,400
Layer 2	Phase III–IV + early V	Transcript, AS-unit/clause, lexical/MWU and pause-location expansion	USD 800
Layer 3	Full Phase V	Final matrix, validation/reporting, archive and UI completion	USD 500
Hosting	Linux VPS	Estimated server hosting	Approx. USD 20/month
O&M	Operations	Monitoring, updates, basic support and small operational fixes	Approx. USD 200/month
AssemblyAI	Monthly allowance	Up to USD 50/month included during active maintenance; overage at actual cost	Included up to USD 50/month

- **Recommended initial contract:** Layer 1 (L1a + L1b) only, fixed implementation fee USD 2,400. Layers 2 and 3 may be activated later by written order or amendment.
- **All-layer implementation ceiling:** USD 3,700 if Layers 1, 2 and 3 are all activated under the frozen scope. Recurring hosting, maintenance and approved third-party usage are separate.
- **Payment terms:** 50% upon activation of each layer and 50% upon acceptance of that layer, unless the parties agree otherwise in writing.
- **Change Request boundary:** routine parameter tuning is included. A written CR applies only to changes in methodology, architecture, contracted corpus type, data schema or deliverable structure. No unlisted third-party charge will be incurred without prior written approval.
- **Quotation validity:** 30 calendar days from the SOW date.

8. Acceptance Criteria

An activated layer is accepted when its agreed deliverables are supplied and the following applicable criteria are met.

1. The original WAV remains the canonical timeline; all interval tiers cover the complete task without gaps or overlaps and pass Praat/headless validation.
2. For the current three-speaker baseline, each P025/P035 TextGrid contains S1, S2, S3, floor, transitions and flags tiers, with only the approved nine labels on speaker tiers.
3. Path B overlap transitions preserve missing FTO explicitly, retain turn-end/start and overlap evidence, and never convert a missing overlap offset to zero.
4. Method manifests record P, L, G, frame size, minimum sounding, minimum overlap, VAD/Praat settings, provider/model provenance, lexicon version, checksums and review state.
5. The Provider supplies a comparison report against the agreed researcher reference using the agreed metrics. Automated output remains a reviewable draft; acceptance does not imply zero manual correction or a dataset-independent accuracy guarantee.
6. Layer 2 is accepted only against the signed transcript, AS-unit/clause, MWU, syllable/repair and tool-output definitions. Unsupported fields are marked pending or excluded, never fabricated.
7. Layer 3 is accepted against the signed final matrix schema, codebook, expected sample rows and agreed validation package.

Validation condition: the corrected calibration recording may be used for tuning, but generalization must be evaluated on a separate researcher-reviewed sample before any corpus-wide accuracy claim is made.

9. Authorization

By signing below, the parties acknowledge and agree to the terms, conditions, scope, method definitions, pricing and dependencies set forth in this Statement of Work.

Accepted and Agreed by Client:

Signature: ____

Name / Title: ____

Date: ____

Accepted and Agreed by Provider:

Signature: ____

Name / Title: ____

Date: ____

0. Appendix & Binding Method Specifications

The following parameters, labels, rules, handoffs and client inputs form the binding technical baseline for this SOW. Routine tuning of the listed adjustable values does not change the method; changing their meaning, ownership or output schema does.

A. Layer and Phase Mapping

Layer	Research phase	Input	Output
L1a	Phase I	Original WAV + metadata	Shared activity, attribution, Stage 1 evidence, speaker review aids, callable floor/label engine
L1b	Phase II	Phase I evidence + P/L/G/Praat config	P025/P035 TextGrids, nine labels, floor, transitions, flags, durations, Path B evidence
L2	Phase III–IV + early V	Reviewed transcript + nine-label timeline + signed research definitions	RAW/TIDY, AS-unit/clause, lexical/MWU, pause-location and rate features
L3	Full Phase V	Accepted L1/L2 artifacts + final schema	Research matrix, workbook, codebook, validation, reports, archive and final UI

B. Initial Method Parameters

Parameter	Initial value	Operational meaning	Status
P	0.25 s and 0.35 s	Minimum pause/gap threshold; independent runs	Adjustable; both defaults preserved
L	1.0 s	Floor-release/shared-lapse threshold	Adjustable by research decision
G	2.0 s	Responsibility-analysis window; does not change shs labeling	Reserved/adjustable
Frame	10 ms	Canonical interaction analysis step	Method parameter
Min sounding	100 ms	Discard shorter acoustic runs as noise	Routine tuning included
Min overlap	100 ms	Minimum simultaneity for ol	Routine tuning included
Overlap band	±100 ms	Associate overlap evidence to transfer boundary	Routine tuning included
Intensity window	200 s	Stable Praat/intensity baseline window	Recorded per run
Gap fill	0 in Phase I	Threshold-neutral provider evidence	P-specific filling only in Phase II
Local RMS VAD	20 ms frame / 10 ms hop	Noise P20; margin 10 dB; floor -55 dB; hysteresis 3 dB; relative 45 dB	Exact derived threshold logged per file

Parameter	Initial value	Operational meaning	Status
Phonation	Include s/f/ol	Exclude bc/op/pf/tr/shs/x by default	Versioned research setting
Route	Path B	Gap-only and transfer flags when overlap offset is not measured	Path A only after reference validation

C. Nine Timeline Labels

Code	Name	Definition	Assignment rule
s	Speech	Lexical speech by the current floor holder.	Counts as phonation; embedded laughter in intelligible speech remains s.
f	Filled hesitation	Nonlexical hesitation by the floor holder: um, uh, er, humming or prolonged vowel.	Counts as phonation and hesitation; holder retains floor.
bc	Backchannel	Short supportive response from a non-holder.	At most 3 words/nonlexical; majority tokens in lexicon; no turn-projecting start; holder continues.
ol	Overlap	Two or more speakers simultaneously vocalise for at least 100 ms, excluding holder plus qualifying bc alone.	Mark every involved speaker ol; counts as phonation; uncertain boundaries are flagged.
op	Own pause	Floor holder is silent but retains the floor.	If same holder resumes, whole silence is op; above L is flagged for review.
pf	Pause while floor held	A non-holder is silent while another speaker holds the floor.	Listening time, not individual hesitation.
tr	Transition gap	Floor is FREE for no more than L before a different speaker takes it.	Mark tr on all speaker tiers; FTO is a separate transition measure.
shs	Shared silence	Floor is FREE and nobody speaks for longer than L, or until task end.	Mark whole lapse shs on all speaker tiers.
x	Unusable/non-speech	Wordless laughter, cough, handling noise or unusable audio.	Formal label; overrides other labels except qualifying soft-chuckle bc; uncertainty is a separate flag.

D. Floor Rules R1–R5

Rule	Name	Binding behavior
R1	Start FREE	The floor is FREE at task start; no holder is assumed before the first turn-taking vocalisation.
R2	Claim floor	The first turn-taking vocalisation from FREE claims the floor for that speaker.
R3	Retain floor	Holder silence and qualifying backchannels do not release or transfer the floor.

Rule	Name	Binding behavior
R4	Competing turn	A genuine non-holder bid transfers only if it survives the holder. Failed bids have no FTO; ambiguous multi-speaker competition is flagged.
R5	Resolve silence	Same holder resumes: op. Another speaker enters within L: tr. Another enters after L, or nobody resumes before task end: shs.

E. Six-Tier TextGrid and Path B Contract

Tier	Praat type	Name	Vocabulary / content
1	IntervalTier	S1	s f bc ol op pf tr shs x
2	IntervalTier	S2	same vocabulary
3	IntervalTier	S3	same vocabulary
4	IntervalTier	floor	S1 S2 S3 FREE
5	TextTier	transitions	signed gap or FTO=NA overlap status
6	IntervalTier	flags	sorted unique review codes joined by |

- All interval tiers begin at 0, end at the canonical task duration, and contain no gaps or overlaps. Adjacent equal intervals are merged.
- Path B qualified overlap (at least 100 ms): fto_sec is missing, status is overlap_present_offset_not_measured, and review is required.
- Path B sub-threshold simultaneity (below 100 ms): no ol label is created; missing FTO and subthreshold_overlap_present_offset_not_measured remain visible in transition evidence and flags.
- Tier 4 floor is the authoritative floor sequence for handoff derivation. Tier 5 is a transition/audit display and must be regenerated when the reviewed floor changes.

F. Required Client Inputs and Activation Conditions

Layer	Required inputs before activation
Layer 1	Original WAVs; task bounds; speaker metadata; confidentiality/provider approval; researcher-corrected six-tier calibration sample; separate validation sample; review strategy.
Layer 2	Reviewed verbatim transcript; filler/repetition/false-start conventions; AS-unit/clause definitions; MWU definition/list; syllable and repair rules; third-party tool versions/settings; gold examples.
Layer 3	Accepted L1/L2 outputs; final matrix schema; codebook; expected sample rows; validation metrics/tolerances; reporting and archive requirements.

Binding dependency: where required definitions, rights, reference data or tool outputs are missing, the affected Layer 2/3 deliverables and schedule will be revised, deferred or removed by mutual written agreement before activation. This does not alter accepted Layer 1 deliverables.

G. Reference Documents

- **Workflow baseline:** MWU Fluency Pipeline — Workflow Atlas, Methodology v2.1.
- **Method baseline:** Technical Appendix — Multilogues, including nine labels and floor rules R1–R5.
- **Calibration reference:** researcher-corrected Multilogue04 P025 six-tier TextGrid and its documented corrections.
- **Validation principle:** calibration and blind validation evidence must remain separate; provider logs establish provenance, not accuracy.